The documentation and process conversion measures necessary to comply with this revision shall be completed by:
31 August 2018.

INCH-POUND

MIL-PRF-19500/565F
31 May 2018
SUPERSEDING
MIL-PRF-19500/565E
1 November 2007

PERFORMANCE SPECIFICATION SHEET

TRANSISTOR, FIELD EFFECT, P-CHANNEL, SILICON,
TYPES 2N6895, 2N6896, 2N6897, AND 2N6898,
QUALITY LEVELS JAN, JANTX, JANTXV, AND JANS

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-PRF-19500.

1. SCOPE

1.1 Scope. This specification covers the performance requirements for P-channel, enhancement-mode, MOSFET, power transistors. Four levels of product assurance (JAN, JANTX, JANTXV and JANS) are provided for each encapsulated device type as specified in MIL-PRF-19500.

1.2 Package outlines. The device packages for the encapsulated device types are as follows: Radial leaded TO-205AF (formerly TO-39) in accordance with figure 1 for type 2N6895; case mount TO-204AA (similar TO-3) in accordance with figure 2 for types 2N6896 and 2N6897; and case mount TO-204AE (formerly TO-3) in accordance with figure 2 for type 2N6898.

1.3 Maximum ratings. Unless otherwise specified, T_A = +25°C.

1.3.1 Ratings applicable to all Part or Identifying Numbers (PIN). $T_J = T_{STG}$ = –55°C to +150°C.

1.3.2 Ratings applicable to individual types.

Type	P_T (1)	P_T	$R_{\theta JC}$	V_{DS}	V_{DG}	V_{GS}	I_{D1} (2) (3)	I_{D2} (2) (3)	I_S	I_{DM}
	T_C = +25°C	T_A = +25°C					T_C = +25°C	T_C = +100°C		
	W	W	°C/W	V dc	V dc	V dc	A dc	A dc	A dc	A(pk)
2N6895	8.33	0.6	15.0	100	100	±20	1.16	0.74	1.16	5
2N6896	60	4	2.083	100	100	±20	6.0	3.8	6.0	20
2N6897	100	4	1.25	100	100	±20	12	7.6	12	30
2N6898	150	4	0.83	100	100	±20	25	15.8	25	60

(1) Derate linearly T_C > +25°C; 2N6895 (0.067 W/°C), 2N6896 (0.48 W/°C), 2N6897 (0.8 W/°C), 2N6898 (1.2 W/°C).
(2) The following formula derives the maximum theoretical I_D limit. I_D is limited by package and internal wires and may be limited by pin diameter:

$$I_D = \sqrt{\frac{T_{JM} - T_C}{(R_{\theta JC}) \times (R_{DS}(on) \text{ at } T_{JM})}}$$

(3) See figure 3, maximum drain current graphs.

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FSC 5961

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1.4 Primary electrical characteristics. Unless otherwise specified, at T_C = +25°C.

Type	Min $V_{(BR)DSS}$ V_{GS} = 0 V I_D = -1.0 mA dc	$V_{GS(th)1}$ $V_{DS} \geq V_{GS}$ I_D = -0.25 mA dc		Max I_{DSS1} V_{GS} = 0 V	Max $r_{DS(on)}$ (1) V_{GS} = -10 V dc	
				V_{DS} = 80 percent of rated V_{DS}	T_J = +25°C at I_{D1}	T_J = +150°C at I_{D2}
	V dc	V dc		µA dc	Ω	Ω
		Min	Max			
2N6895	-100	-2.0	-4.0	-1.0	3.65	6.15
2N6896	-100	-2.0	-4.0	-1.0	0.6	1.67
2N6897	-100	-2.0	-4.0	-1.0	0.3	0.69
2N6898	-100	-2.0	-4.0	-1.0	0.2	0.24

(1) Pulsed (see 4.5.1).

1.5 Part or Identifying Number (PIN). The PIN is in accordance with MIL-PRF-19500, and as specified herein. See 6.4 for PIN construction example and 6.5 for a list of available PINs.

1.5.1 JAN certification mark and quality level. The quality level designators for encapsulated devices that are applicable for this specification sheet from the lowest to the highest level are as follows: "JAN", "JANTX", "JANTXV", and "JANS".

1.5.2 Device type. The designation system for the devices covered by this specification sheet is as follows.

1.5.2.1 First number and first letter symbols. The devices of this specification sheet use the first number and letter symbols "2N".

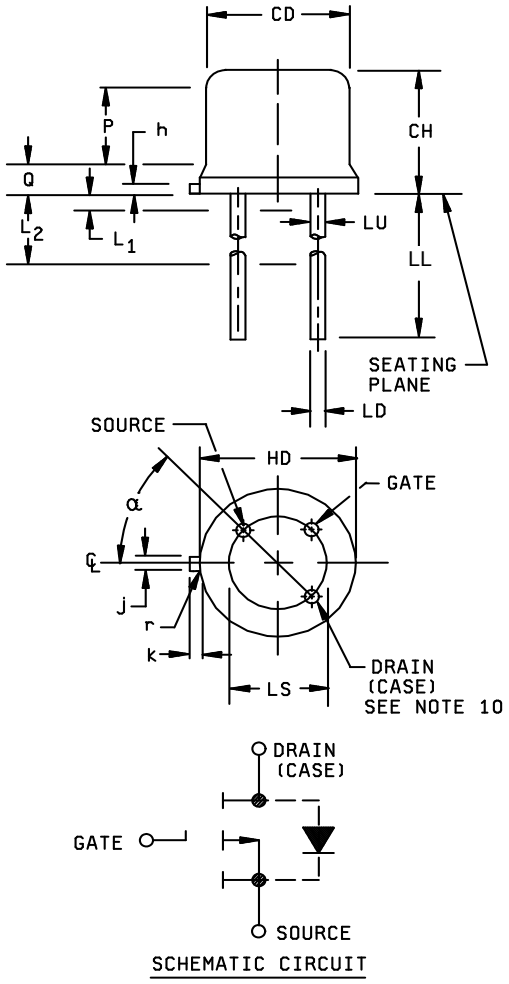
1.5.2.2 Second number symbols. The second number symbols for the devices covered by this specification sheet are as follows: "6895", "6896", "6897", and "6898".

1.5.3 Suffix symbols. Suffix symbols are not applicable to this specification sheet.

1.5.4 Lead finish. The lead finishes applicable to this specification sheet are listed on QPDSIS-19500.

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Ltr	Dimensions				Notes
	Inches		Millimeters		
	Min	Max	Min	Max	
CD	.305	.355	7.75	8.51	2
CH	.160	.180	4.07	4.57	
HD	.335	.370	8.51	9.40	
h	.009	.041	0.23	1.04	
j	.028	.034	0.71	0.86	3
k	.029	.045	0.74	1.14	4
LD	.016	.021	0.41	0.53	5, 6
LL	.500	.750	12.70	19.05	5, 6
LS	.200 TP		5.08 TP		7
LU	.016	.019	0.41	0.48	5, 6
L_1		.050		1.27	5, 6
L_2	.250		6.35		5, 6
P	.100		2.54		2
Q		.050		1.27	8
r		.010		0.25	9
α	45° TP		45° TP		7



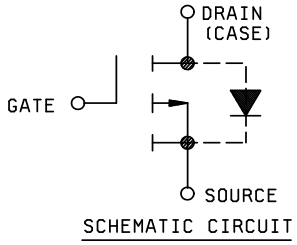
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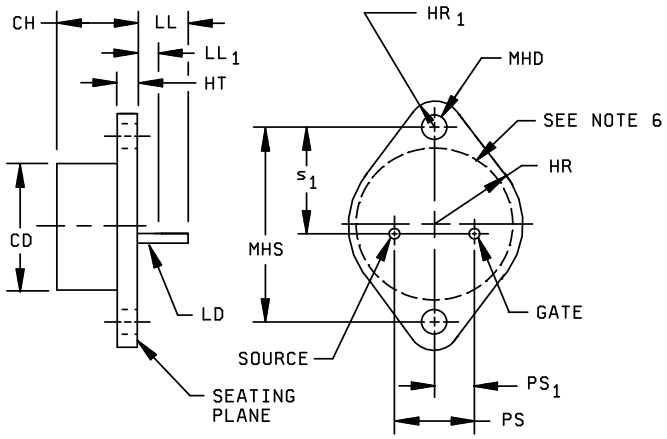
1. Dimensions are in inches. Millimeters are given for general information only.
2. Dimension CD shall not vary more than .010 inch (0.25 mm) in zone P. This zone is controlled for automatic handling.
3. Beyond radius (r) maximum, dimension j shall be held for a minimum length of .011 inch (0.28 mm).
4. Dimension k measured from maximum HD.
5. Dimension LD applies between L_2 and L minimum. Dimension LU applies between L_1 and L_2. Diameter is uncontrolled in L_1 and beyond dimension LL minimum.
6. All three leads.
7. Leads at gauge plane .054 +.001, –.000 inch (1.37 +0.03, –0.00 mm) below seating plane shall be within .007 inch (0.18 mm) radius of true position (TP) at maximum material condition (MMC) relative to tab at MMC. The device may be measured by direct methods.
8. Outline in this zone is not controlled.
9. Radius (r) applies to both inside corners of tab.
10. Drain is electrically connected to the case.
11. In accordance with ASME Y14.5M, diameters are equivalent to ϕx symbology.

FIGURE 1. Physical dimensions for 2N6895 (modified TO-205AF).

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Ltr	Dimensions				Notes
	Inches		Millimeters		
	Min	Max	Min	Max	
CD		.875		22.23	
CH	.250	.360	6.35	9.14	
HR	.495	.525	12.57	13.34	
HR1	.131	.188	3.33	4.78	
HT	.060	.135	1.52	3.43	
LD	.038	.043	0.97	1.09	2
	.057	.063	1.45	1.60	3
LL	.312	.500	7.92	12.70	
LL_1		.050		1.27	
MHD	.151	.161	3.84	4.09	
MHS	1.177	1.197	29.90	30.40	
PS	.420	.440	10.67	11.18	4, 6
PS_1	.205	.225	5.21	5.72	4, 6
S_1	.655	.675	16.64	17.15	





NOTES:

1. Dimensions are in inches. Millimeters are given for general information only.
2. Dimension LD for device types 2N6896 and 2N6897.
3. Dimension LD for device type 2N6898.
4. These dimensions should be measured at points .050 inch (1.27 mm) and .055 inch (1.40 mm) below seating plane. Measurement will be made at the seating plane.
5. The seating plane of the header shall be flat within .001 inch (0.03 mm) concave to .004 inch (0.10 mm) convex inside a .930 inch (23.62 mm) diameter circle on the center of the header and flat within .001 inch (0.03 mm) concave to .006 inch (0.15 mm) convex overall.
6. Mounting holes shall be deburred on the seating plane side.
7. Drain is electrically connected to the case.
8. In accordance with ASME Y14.5M, diameters are equivalent to ϕx symbology.

FIGURE 2. Physical dimensions of transistor 2N6896 and 2N6897 (TO-204AA), and 2N6898 (TO-204AE).

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2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

DEPARTMENT OF DEFENSE SPECIFICATIONS

MIL-PRF-19500 – Semiconductor Devices, General Specification for.

DEPARTMENT OF DEFENSE STANDARDS

MIL-STD-750 – Test Methods For Semiconductor Devices.

(Copies of these documents are available online at http://quicksearch.dla.mil.)

2.3 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 General. The individual item requirements shall be as specified in MIL-PRF-19500 and as modified herein.

3.2 Qualification. Devices furnished under this specification shall be products that are manufactured by a manufacturer authorized by the qualifying activity for listing on the applicable qualified manufacturers list before contract award (see 4.2 and 6.3).

3.3 Abbreviations, symbols, and definitions. Abbreviations, symbols, and definitions used herein shall be as specified in MIL-PRF-19500 and as follows.

nC - nano Coulomb

3.4 Interface and physical dimensions. Interface and physical dimensions shall be as specified in MIL-PRF-19500, and on figure 1, TO-205AF (formerly TO-39) for 2N6895, and figure 2, TO-204AA for 2N6896 and 2N6897; and TO-204AE for 2N6898 (formerly TO-3).

3.4.1 Lead finish. Lead finish shall be solderable in accordance with MIL-PRF-19500, MIL-STD-750, and herein. Where a choice of lead finish is desired, it shall be specified in the acquisition document (see 6.2).

3.4.2 Pin-out. The pin-out of the device shall be as shown on figures 1 and 2. For all package configurations, the drain shall be electrically connected to the case.

3.4.3 Internal construction. Multiple chip construction shall not be permitted.

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3.5 Electrostatic discharge sensitive (ESDS). The devices covered by this specification sheet have been classified as ESDS. The devices shall be handled in accordance with the ESD program established to comply with the requirements MIL-PRF-19500 to avoid damage due to the accumulation of static charge. The following handling practices shall be followed:

a. Devices shall be handled on benches with conductive handling devices.

b. Ground test equipment, tools, and personnel handling devices.

c. Do not handle devices by the leads.

d. Store devices in conductive foam or carriers.

e. Avoid use of plastic, rubber, or silk in MOS areas.

f. Maintain relative humidity above 50 percent if practical.

g. Care shall be exercised, during test and troubleshooting, to apply not more than maximum rated voltage to any lead.

h. Gate must be terminated to source, R ≤ 100 kΩ, whenever bias voltage is to be applied drain to source.

3.6 Marking. Marking shall be in accordance with MIL-PRF-19500. At the option of the manufacturer, marking of country of origin may be omitted from the body of the transistor, but shall be retained on the initial container.

3.7 Electrical performance characteristics. Unless otherwise specified herein, the electrical performance characteristics are as specified in 1.3, 1.4, and table I.

3.8 Electrical test requirements. The electrical test requirements shall be as specified in table I.

3.9 Workmanship. Semiconductor devices shall be processed in such a manner as to be uniform in quality and shall be free from other defects that will affect life, serviceability, or appearance.

4. VERIFICATION

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. Qualification inspection (see 4.2).

b. Screening (see 4.3).

c. Conformance inspection (see 4.4 and table I).

4.2 Qualification inspection. Qualification inspection shall be in accordance with MIL-PRF-19500 and as specified herein.

4.2.1 Group E qualification. Group E inspection shall be performed for qualification or re-qualification only. In case qualification was awarded to a prior revision of the specification sheet that did not request the performance of table II tests, the tests specified in table II herein that were not performed in the prior revision shall be performed on the first inspection lot of this revision to maintain qualification.

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4.3 Screening (quality levels JANS, JANTX, and JANTXV only). Screening shall be in accordance with table E-IV of MIL-PRF-19500 and as specified herein. The following measurements shall be made in accordance with table I herein. Devices that exceed the limits of table I herein shall not be acceptable.

Screen (1) (2)	Measurement	
	JANS level	JANTX and JANTXV levels
(3)	Gate stress test (see 4.3.1)	Gate stress test (see 4.3.1)
(3)	Method 3470 of MIL-STD-750, (see 4.3.2)	Method 3470 of MIL-STD-750, (see 4.3.2)
(3) 3c	Method 3161 of MIL-STD-750, (see 4.3.3)	Method 3161 of MIL-STD-750, (see 4.3.3)
9	Subgroup 2 of table I herein.	Not applicable.
10	Method 1042 of MIL-STD-750, test condition B.	Method 1042 of MIL-STD-750, test condition B.
11	Subgroup 2 of table I herein; ΔI_{GGSF1} = ±20 nA dc or ±100 percent of initial value, whichever is greater. ΔI_{GGSR1} = ±20 nA dc or ±100 percent of initial value, whichever is greater. ΔI_{DSS1} = ±0.2 µA dc or ±100 percent of initial value, whichever is greater.	Subgroup 2 of table I herein.
12	Method 1042 of MIL-STD-750, test condition A and test condition C (see 4.3.4).	Method 1042 of MIL-STD-750, test condition A.
13	Subgroups 2 and 3 of table I herein; ΔI_{GGSF1} = ±20 nA dc or ±100 percent of initial value, whichever is greater. ΔI_{GGSR1} = ±20 nA dc or ±100 percent of initial value, whichever is greater. ΔI_{DSS1} = ±.2 µA dc or ±100 percent of initial value, whichever is greater. $\Delta r_{DS(on)1}$ = ±20 percent of initial value. $\Delta V_{GS(th)}$ = ±20 percent of initial value.	Subgroup 2 of table I herein; ΔI_{GGSF1} = ±20 nA dc or ±100 percent of initial value, whichever is greater. ΔI_{GGSR1} = ±20 nA dc or ±100 percent of initial value, whichever is greater. ΔI_{DSS1} = ±.2 µA dc or ±100 percent of initial value, whichever is greater. $\Delta r_{DS(on)1}$ = ±20 percent of initial value. $\Delta V_{GS(th)}$ = ±20 percent of initial value.

(1) At the end of the test program, I_{GGSF1}, I_{GGSR1}, I_{DSS1} are measured.
(2) An out-of-family program to characterize I_{GGSF1}, I_{GGSR1}, I_{DSS1}, $V_{GS(th)1}$ and $r_{DS(ON)1}$ shall be invoked.
(3) Thermal impedance shall be performed any time after sealing provided temperature cycling is performed in accordance with MIL-PRF-19500, screen 3a; JANTX and JANTXV levels do not need to be repeated in screening requirements.

4.3.1 Gate stress test. Apply V_{GS} = -30 V minimum for t = 250 µs minimum.

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4.3.2 Unclamped inductive switching. The unclamped inductive switching test shall be performed in accordance with test method 3470 of MIL-STD-750. The following details shall apply:

a. Peak current (I_D)

2N6895	1 A
2N6896	2.7 A
2N6897	5 A
2N6898	5.9 A

b. Peak gate voltage (V_{GS})..... 10 V.

c. Gate to source resistor (R_{GS}) ..... $25\Omega \leq R_{GS} \leq 200\Omega$.

d. Initial case temperature (T_C) ..... +25°C, +10°C, –5°C.

e. Inductance (L)..... 100 µH ±10 percent.

f. Number of pulses to be applied ..... 1 pulse minimum.

g. Pulse repetition rate..... None.

4.3.3 Thermal impedance. The thermal impedance measurements shall be performed in accordance with method 3161 of MIL-STD-750 using the guidelines in that method for determining I_M, I_H, t_H, t_{SW}, (and V_H where appropriate). Measurement delay time (t_{MD}) = 70 µs max. See table II, subgroup 4 herein.

4.3.4 Power burn-in. The power burn-in of devices shall be performed in accordance with condition C of test method 1042 of MIL-STD-750. The power burn-in conditions shall be as follows: T_A = +25°C, –5°C, +10°C, V_{DS} = 10 V min.; I_D adjusted to meet a junction temperature of +140°C, –5°C, +10°C, t = 240 hours.

4.4 Conformance inspection. Conformance inspection shall be in accordance with MIL-PRF-19500. Alternate flow is allowed for conformance inspection in accordance with figure 4 of MIL-PRF-19500.

4.4.1 Group A inspection. Group A inspection shall be conducted in accordance with MIL-PRF-19500 and table I herein.

4.4.2 Group B inspection. Group B inspection shall be conducted in accordance with the conditions specified for subgroup testing in table E-VIA (JANS) and table E-VIB (JAN, JANTX, and JANTXV) of MIL-PRF-19500 and herein.

4.4.2.1 Quality level JANS.

Subgroup	Method	Conditions
B3	1051	Test condition G.
B3	2077	SEM.
B4	1042	Test condition D. The heating cycle shall be 1 minute minimum. 2N6895, V_{DS} = –10 V dc, P_T = 0.6 W at T_A = +25°C ±3°C. 2N6896, 2N6897, 2N6898, V_{DS} = –20 V dc, P_T = 4 W at T_A = +25°C ±3°C.
B5	1042	Accelerated steady-state operation life; test condition C; T_A = +25°C, –5°C, +10°C, V_{DS} = 10 V min.; I_D adjusted to meet a junction temperature of +140°C, –0°C, +10°C, t = 240 hours.
B5	2037	Bond strength, test condition D.

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4.4.2.2 Quality levels JAN, JANTX, and JANTXV.

Subgroup	Method	Conditions
B2	1051	Test condition G, 25 cycles.
B3	1042	Test condition D, 2,000 cycles. The heating cycle shall be 1 minute minimum.

4.4.3 Group C inspection. Group C inspection shall be conducted in accordance with the conditions specified for subgroup testing in table E-VII of MIL-PRF-19500 and as follows.

Subgroup	Method	Conditions
C2	2036	Test condition E for TO-205AF (figure 1); Test condition A, 10 lbs (4.5.4 Kg), 15 s for TO-204AA and TO-204AE (figure 2).
C5	3161	See 4.3.3, $R_{\theta JC}$ = see 1.3.
C6	1042	Test condition D, 6,000 cycles. The heating cycle shall be 1 minute minimum.

4.4.4 Group E inspection. Group E inspection shall be conducted in accordance with the conditions specified for subgroup testing in table E-IX of MIL-PRF-19500 and as specified in table II herein. Electrical measurements shall be in accordance with the inspections of table I, subgroup 2 herein.

4.5 Methods of inspection. Methods of inspection shall be as specified in the appropriate tables and as follows.

4.5.1 Pulse measurements. Conditions for pulse measurement shall be as specified in section 4 of MIL-STD-750.

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TABLE I. Group A inspection.

Inspection 1/	MIL-STD-750		Symbol	Limits		Unit
	Method	Conditions		Min	Max	
Subgroup 1						
Visual and mechanical inspection	2071					
Subgroup 2						
Thermal impedance 2/	3161	See 4.3.3	$Z_{\theta JX}$			°C/W
Breakdown voltage, drain-to-source	3407	Bias condition C, V_{GS} = 0 V; I_D = –1.0 mA dc	$V_{(BR)DSS}$	–100		V dc
Gate-to-source voltage (threshold)	3403	$V_{DS} \geq V_{GS}$; I_D = –0.25 mA dc	$V_{GS(th)1}$	–2.0	–4.0	V dc
Gate current (forward with all other terminals short-circuited to source)	3411	Bias condition C; V_{DS} = 0 V; V_{GS} = +20 V dc	I_{GSSF1}		+100	nA dc
Gate current (reverse with all other terminals short-circuited to source)	3411	Bias condition C; V_{DS} = 0 V; V_{GS} = –20 V dc	I_{GSSR1}		–100	nA dc
Drain current	3413	Bias condition C; V_{GS} = 0; V_{DS} = –80 V	I_{DSS1}		–1.0	µA dc
Static drain to source on-state resistance	3421	Condition A; pulsed (see 4.5.1); V_{GS} = –10 V dc;	$r_{DS(on)1}$			
2N6895		I_D = –0.74 A dc			3.65	Ω
2N6896		I_D = –3.8 A dc			0.6	Ω
2N6897		I_D = –7.6 A dc			0.3	Ω
2N6898		I_D = –15.8 A dc			0.2	Ω
Drain to source on-state voltage	3405	Condition A; pulsed (see 4.5.1); V_{GS} = 10 V dc	$V_{DS(on)1}$			
2N6895		I_D = –1.16 A dc			–6.0	V
2N6896		I_D = –6.0 A dc			–6.0	V
2N6897		I_D = –12.0 A dc			–4.8	V
2N6898		I_D = –25.0 A dc			–6.0	V
Forward voltage (source drain diode)	4011	Condition B; pulsed (see 4.5.1); V_{GS} = 0 V	V_{SD}			
2N6895		I_D = –1.16 A dc		–0.8	–1.6	V
2N6896		I_D = –6.0 A dc		–0.8	–1.6	V
2N6897		I_D = –12.0 A dc		–0.8	–1.6	V
2N6898		I_D = –25.0 A dc		–0.8	–1.6	V

See footnotes at end of table.

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TABLE I. Group A inspection - Continued.

Inspection 1/	MIL-STD-750		Symbol	Limits		Unit
	Method	Conditions		Min	Max	
Subgroup 2 - Continued.						
Forward transconductance (pulsed dc method)	3475	Pulsed (see 4.5.1), I_D = rated I_{D2} (see 1.3)	g_{fs}			
2N6895				0.2		s
2N6896				1.0		s
2N6897				2.0		s
2N6898				4.0		s
Subgroup 3						
High temperature operation:		$T_C = T_J = +125°C$				
Gate-to-source voltage (threshold)	3403	$V_{DS} \geq V_{GS}$, I_D = –0.25 mA dc	$V_{GS(th)2}$	–1.0		V dc
Gate current (forward and reverse with all other terminals short-circuited to source)	3411	Bias condition C, V_{DS} = 0 V; V_{GS} = +20 V dc and -20 V dc	I_{GSS2}		±200	nA dc
Drain current	3413	Bias condition C, V_{GS} = 0 V, V_{DS} = –80 V	I_{DSS2}		–50	µA dc
Static drain-to-source on-state resistance	3421	Bias condition A, V_{GS} = –10 V dc, pulsed (see 4.5.1)	$r_{DS(on)2}$			
2N6895		I_D = –0.74 A dc			5.66	Ω
2N6896		I_D = –3.8 A dc			0.96	Ω
2N6897		I_D = –7.6 A dc			0.465	Ω
2N6898		I_D = –15.8 A dc			0.24	Ω
Low temperature operation:		$T_C = T_J = –55°C$				
Gate-to-source voltage (threshold)	3403	$V_{DS} \geq V_{GS}$, I_D = –0.25 mA	$V_{GS(th)3}$		–5.0	V dc
Subgroup 4						
Switching time test	3472	I_D = rated I_{D2} (see 1.3 and below); V_{DD} = 50 percent of rated V_{DS} (see 1.3 and below), V_{GS} = 10 V dc; R_{GS} = 15 Ω, R_{gen} = 15 Ω;				
Turn-on delay time		V_{DD} = –50 V dc	$t_{d(on)}$			
2N6895		I_D = –0.74 A dc			25	ns
2N6896		I_D = –3.8 A dc			60	ns
2N6897		I_D = –7.6 A dc			60	ns
2N6898		I_D = –15.8 A dc			50	ns

See footnotes at end of table.

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TABLE I. Group A inspection - Continued.

Inspection 1/	MIL-STD-750		Symbol	Limits		Unit
	Method	Conditions		Min	Max	
Subgroup 4 - Continued.						
Rise time		V_{DD} = –50 V dc	t_r			
2N6895		I_D = –0.74 A dc			45	ns
2N6896		I_D = –3.8 A dc			100	ns
2N6897		I_D = –7.6 A dc			175	ns
2N6898		I_D = –15.8 A dc			250	ns
Turn-off delay time		V_{DD} = –50 V dc	$t_{d(off)}$			
2N6895		I_D = –0.74 A dc			45	ns
2N6896		I_D = –3.8 A dc			150	ns
2N6897		I_D = –7.6 A dc			275	ns
2N6898		I_D = –15.8 A dc			400	ns
Fall time		V_{DD} = –50 V dc	t_f			
2N6895		I_D = –0.74 A dc			50	ns
2N6896		I_D = –3.8 A dc			100	ns
2N6897		I_D = –7.6 A dc			175	ns
2N6898		I_D = –15.8 A dc			250	ns
Subgroup 5						
Safe operating area	3474	See figure 4.				
High voltage test		V_{DS} = 80 percent of rated V_{DS} (see 1.3)				
Electrical measurements		See table I, subgroup 2				
Subgroup 6						
Not applicable						
Subgroup 7						
Gate charge	3471	Condition A or B				
Test 1						
On-state gate charge			$Q_{g(on)}$			
2N6895		I_D = –0.74 A dc		2.2	4.7	nC
2N6896		I_D = –3.8 A dc		13	24	nC
2N6897		I_D = –7.6 A dc		31	58	nC
2N6898		I_D = –15.8 A dc		50	117	nC

See footnotes at end of table.

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TABLE I. Group A inspection - Continued.

Inspection 1/	MIL-STD-750		Symbol	Limits		Unit
	Method	Conditions		Min	Max	
Subgroup 7 - Continued.						
Test 2						
Gate to source charge			Q_{gs}			
2N6895				0.4	1.2	nC
2N6896				1.1	5.5	nC
2N6897				3	13	nC
2N6898				6	25	nC
Test 3						
Gate to drain charge			Q_{gd}			
2N6895				0.9	2.9	nC
2N6896				5.5	14.5	nC
2N6897				14	36	nC
2N6898				26	69	nC
Reverse recovery time	3473	Condition A; $V_{DD} \leq 30$ V; di/dt = 100A/µs; I_F = 4 A	t_{rr}			
2N6895					340	ns
2N6896					375	ns
2N6897					500	ns
2N6898					750	ns

1/ For sampling plan, see MIL-PRF-19500.
2/ For end point electrical measurements, this test is required for the following subgroups:
Group B, subgroups 2 and 3 (JANTXV).
Group B, subgroups 3 and 4 (JANS).
Group C, subgroup 2 and 6.
Group E, subgroup 1.

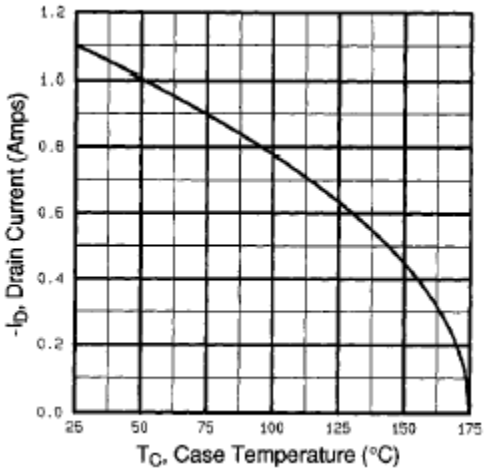
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TABLE II. Group E inspection (all quality levels) for qualification or re-qualification only.

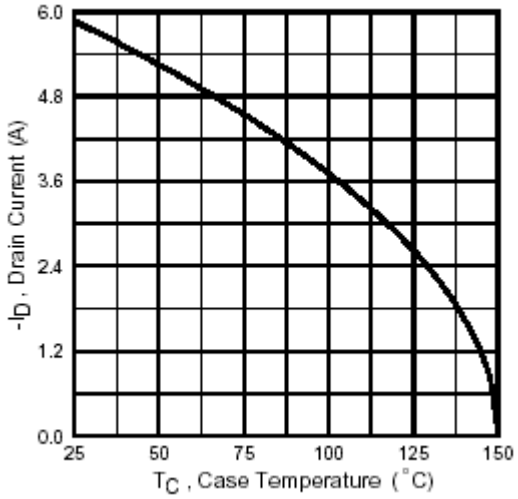
Inspection	MIL-STD-750		Sample plan
	Method	Conditions	
Subgroup 1			45 devices c = 0
Temperature cycling	1051	Test condition G.	
Hermetic seal Fine leak Gross leak	1071		
Electrical measurements		See table I, subgroup 2.	
Subgroup 2 1/			45 devices c = 0
Steady-state gate bias	1042	Condition B, 1,000 hours.	
Electrical measurements		See table I, subgroup 2.	
Steady-state reverse bias	1042	Condition A, 1,000 hours.	
Electrical measurements		See table I, subgroup 2.	
Subgroup 4			Sample size N/A
Thermal impedance curves		See MIL-PRF-19500.	
Subgroup 6			
ESD	1020		11 devices
Subgroup 10			22 devices c = 0
Commutating diode for safe operating area test procedure for measuring dv/dt during reverse recovery of power MOSFET transistors or insulated gate bipolar transistors	3476	Test conditions shall be derived by the manufacturer	

1/ A separate sample may be pulled for each test condition.

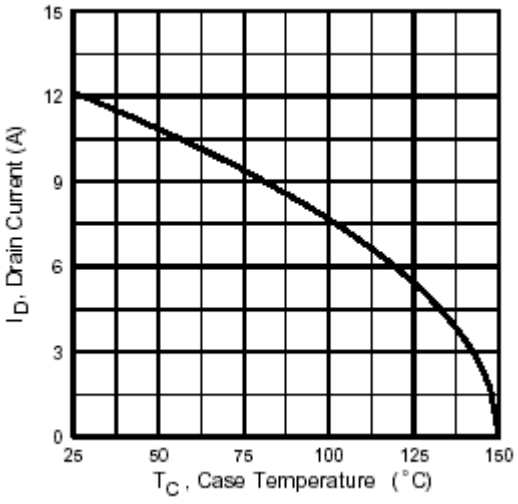
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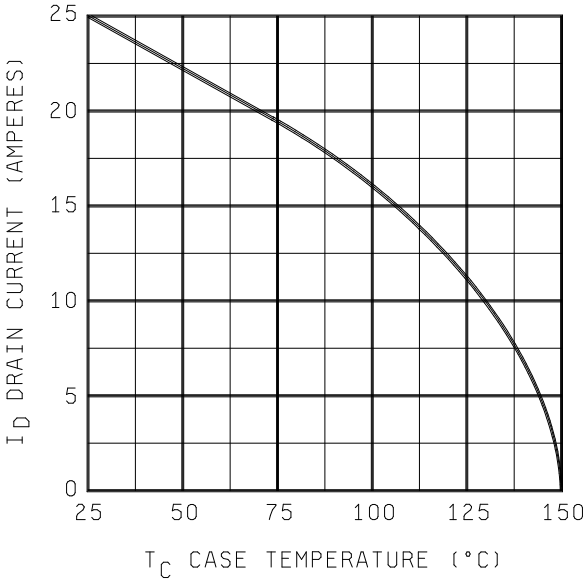
2N6895



2N6896



2N6897



2N6898

FIGURE 3. Maximum drain current vs case temperature graphs.

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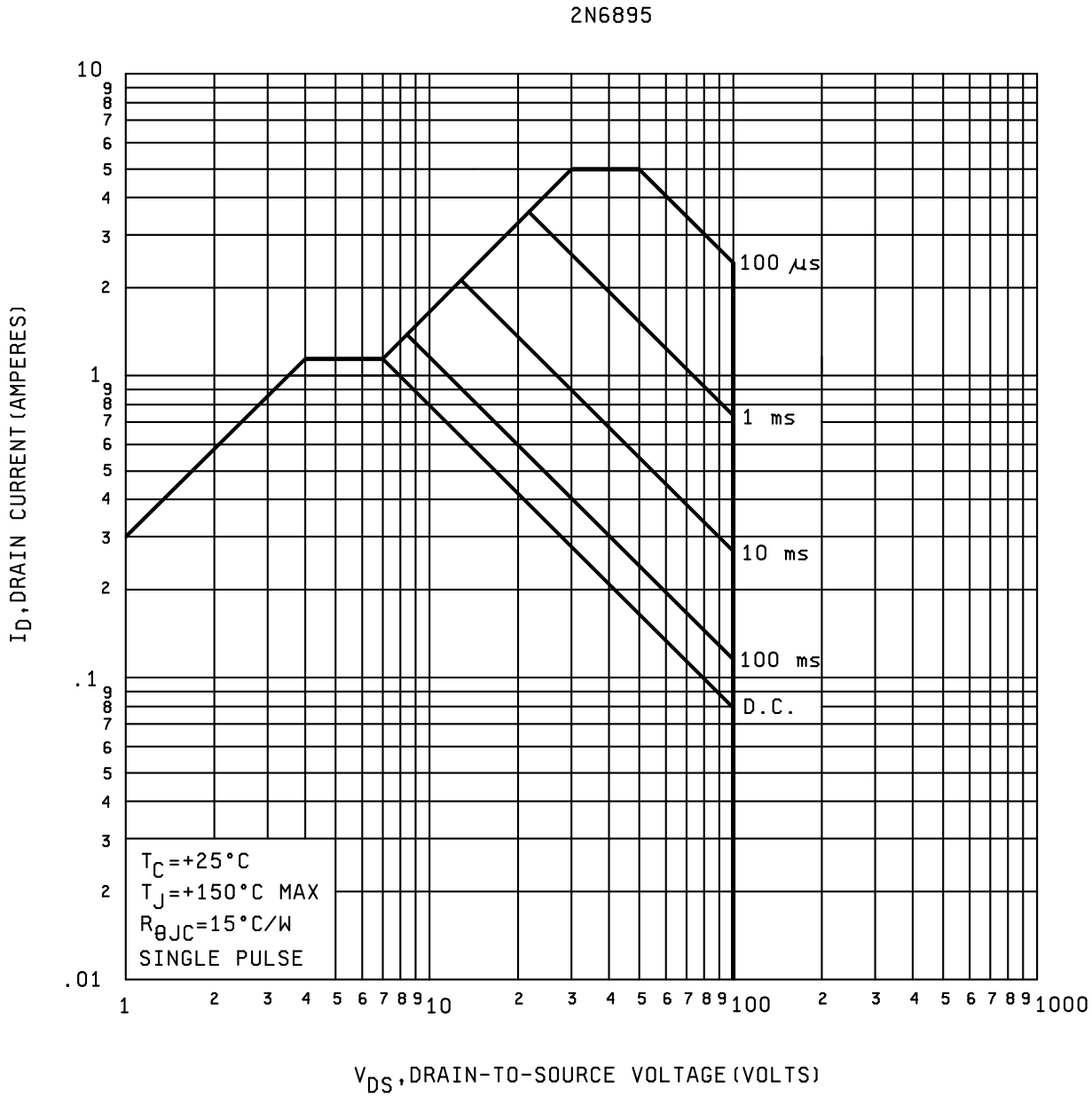


FIGURE 4. Maximum safe operating area.

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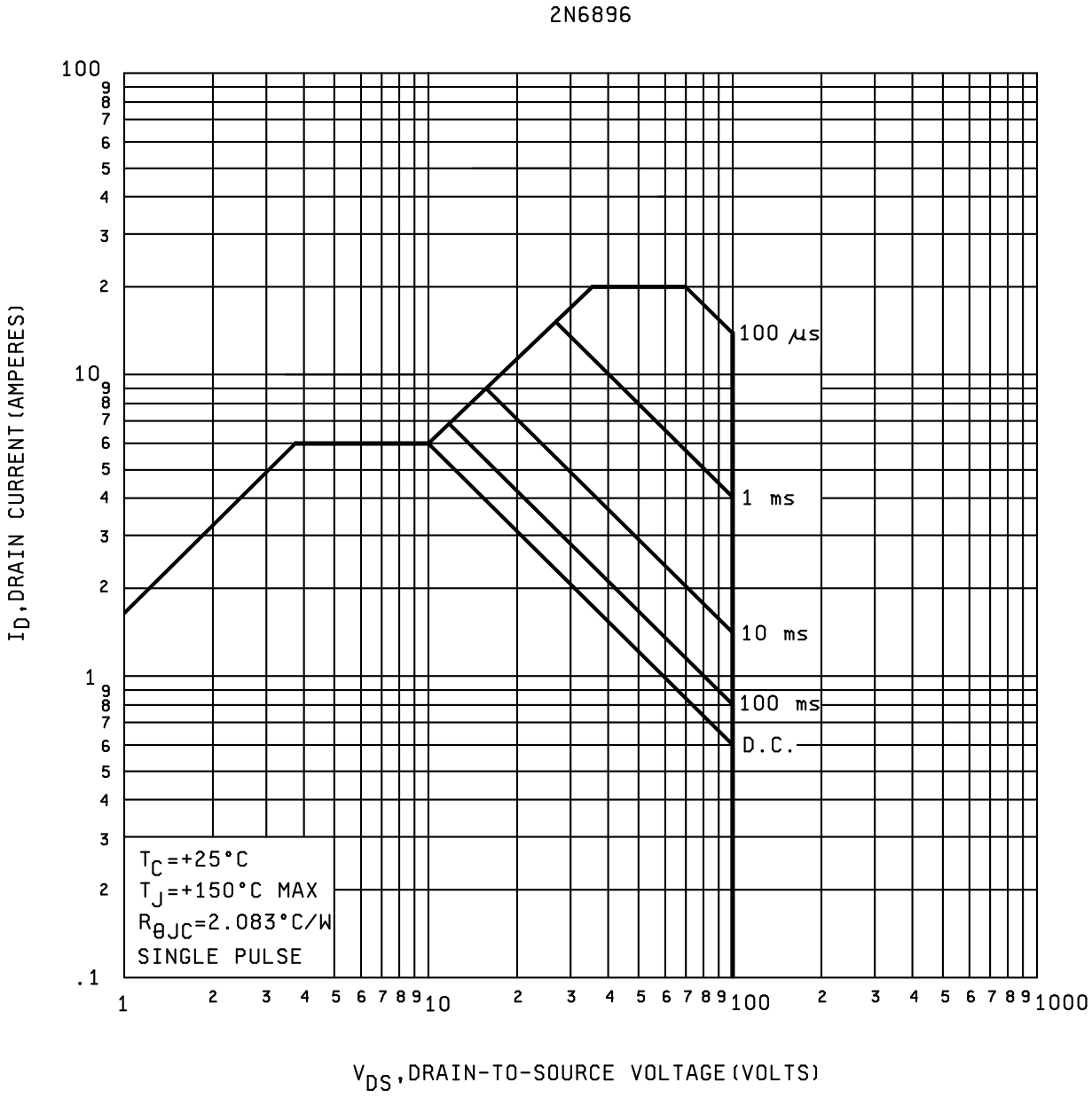


FIGURE 4. Maximum safe operating area – Continued.

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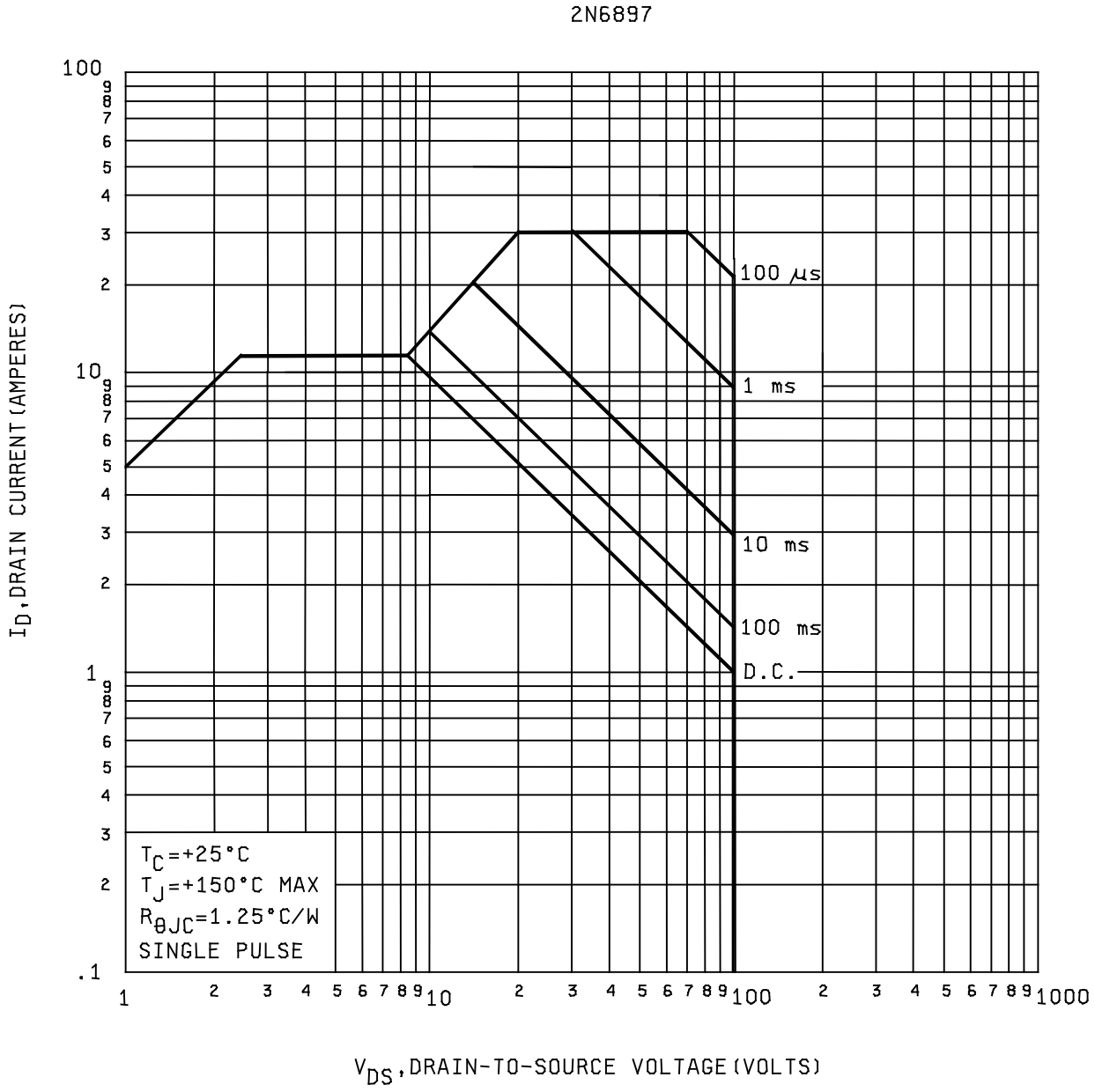


FIGURE 4. Maximum safe operating area – Continued.

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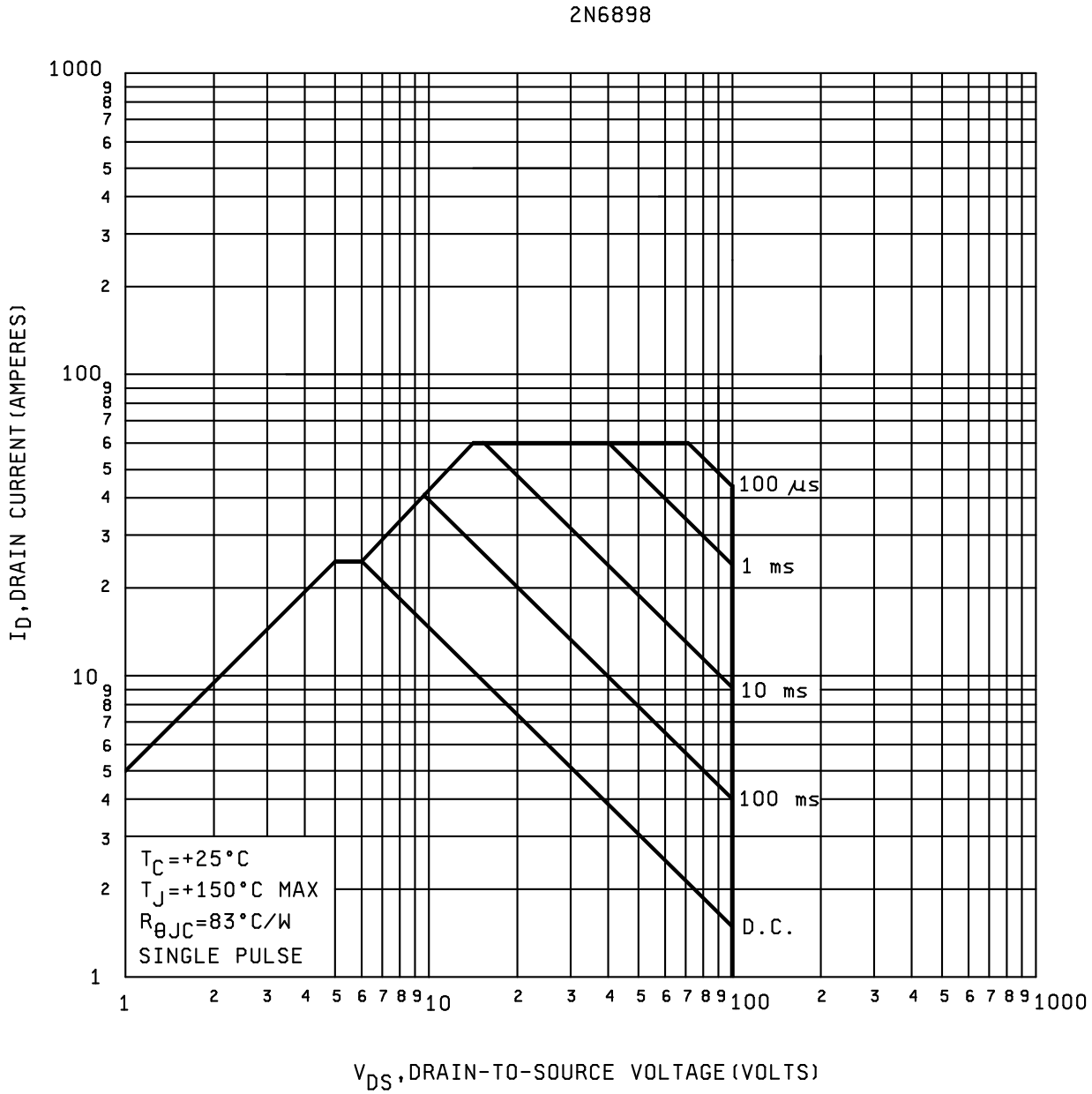


FIGURE 4. Maximum safe operating area – Continued.

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5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Service or Defense Agency, or within the Military Service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory. The notes specified in MIL-PRF-19500 are applicable to this specification.)

6.1 Intended use. Semiconductors conforming to this specification are intended for original equipment design applications and logistic support of existing equipment.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.

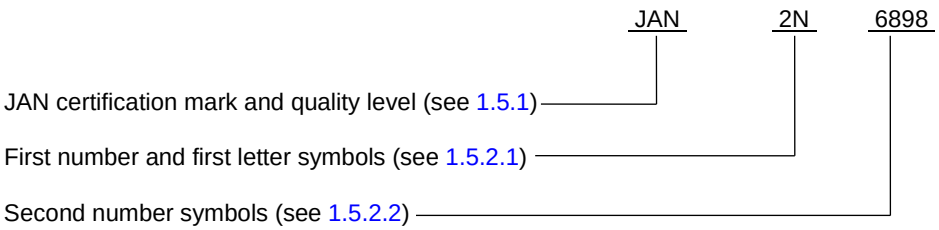
b. Packaging requirements (see 5.1).

c. Lead finish (see 3.4.1).

d. The complete PIN, see 1.5 and 6.5.

6.3 Qualification. With respect to products requiring qualification, awards will be made only for products which are, at the time of award of contract, qualified for inclusion in Qualified Manufacturers List (QML-19500) whether or not such products have actually been so listed by that date. The attention of the contractors is called to these requirements, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or orders for the products covered by this specification. Information pertaining to qualification of products may be obtained from DLA Land and Maritime, ATTN: VQE, P.O. Box 3990, Columbus, OH 43218-3990 or e-mail vqe.chief@dla.mil. An online listing of products qualified to this specification may be found in the Qualified Products Database (QPD) at http://qpldocs.dla.mil.

6.4 PIN construction example. The PINs for encapsulated devices are constructed using the following form.



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6.5 List of PINs. The following is a list of possible PINs available for encapsulated devices covered by this specification sheet.

JAN quality level	JANTX quality level	JANTXV quality level	JANS quality level
JAN2N6895	JANTX2N6895	JANTXV2N6895	JANS2N6895
JAN2N6896	JANTX2N6896	JANTXV2N6896	JANS2N6896
JAN2N6897	JANTX2N6897	JANTXV2N6897	JANS2N6897
JAN2N6898	JANTX2N6898	JANTXV2N6898	JANS2N6898

6.6 Lead material. The original issue of this specification through MIL-PRF-19500/565E (1 November 2007) specified a lead material. Because of the performance format of this document, a lead material is no longer specified.

6.7 Request for new types and configurations. Requests for new device types or configurations for inclusions in this specification sheet should be submitted to: DLA Land and Maritime, ATTN: VAC, Post Office Box 3990, Columbus, OH 43218-3990 or by electronic mail at Semiconductor@dla.mil or by facsimile (614) 693-1642 or DSN 850-6939.

6.8 Changes from previous issue. The margins of this specification are marked with vertical lines to indicate where changes from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

Custodians:
Army – CR
Navy – EC
Air Force – 85
NASA – NA
DLA – CC

Review activities:
Army – AR, MI, SM
Navy – AS, MC
Air Force – 19, 99

Preparing activity:
DLA – CC

(Project 5961–2018–028)

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